

Chapter 1

Introduction

Online booking is the process of scheduling an appointment or reserving a service via an online platform. Many businesses have adopted these systems to manage and organize appointments and reservations, and to provide clients with a convenient way to access their services. According to Maglione (2024), a booking system can help you do way more than just passively accept bookings and payments online. Reservation technology (res-tech) has evolved to the point that it's become a central hub for managing every aspect of your business, from marketing to distribution to operations.

Despite the availability of modern technology, many nail businesses continue to rely on manual booking methods, such as in-person walk-ins, direct messaging through social media, phone calls, and SMS. This method is generally recorded in physical logs or digital calendars, with staff responsible for managing each reservation directly, which is time-consuming and prone to errors. This manual method may lead to delayed responses, double bookings, missed appointments, limited walk-in availability, and clients can cancel their booked reservations at any time. Aside from the booking challenges, they also face difficulty communicating with their clients about choosing suitable nail designs. The client sends their nail design to the staff online, which can

sometimes lead to confusion because communication can break down when the client keeps changing their mind.

Innovative technologies like Artificial Intelligence (AI) enable computers and machines to perform tasks that normally require human thinking. It can follow instructions and generate outputs based on what people want or ask. According to Stryker and Kavlakoglu (2024), AI can understand and respond to human language. They can make detailed recommendations to users and experts. They can act independently, replacing the need for human intelligence or intervention.

In the case of Tips by Nadine, both appointment booking and nail design selection are handled manually. There is no centralized platform where clients can check available nail service dates, view nail designs and their prices, or upload their reference nail design. This situation can pose various challenges for businesses.

To address these issues, this study aims to develop a Web-Based Client Booking System with AI-Assisted Nail Design Recommendation for Tips by Nadine. The proposed system will allow clients to check available dates, book appointments online, view nail designs with corresponding prices, upload reference photos, and receive AI-generated nail design recommendations. In

addition, the system will send email notifications to remind both clients and the owner of upcoming appointments.

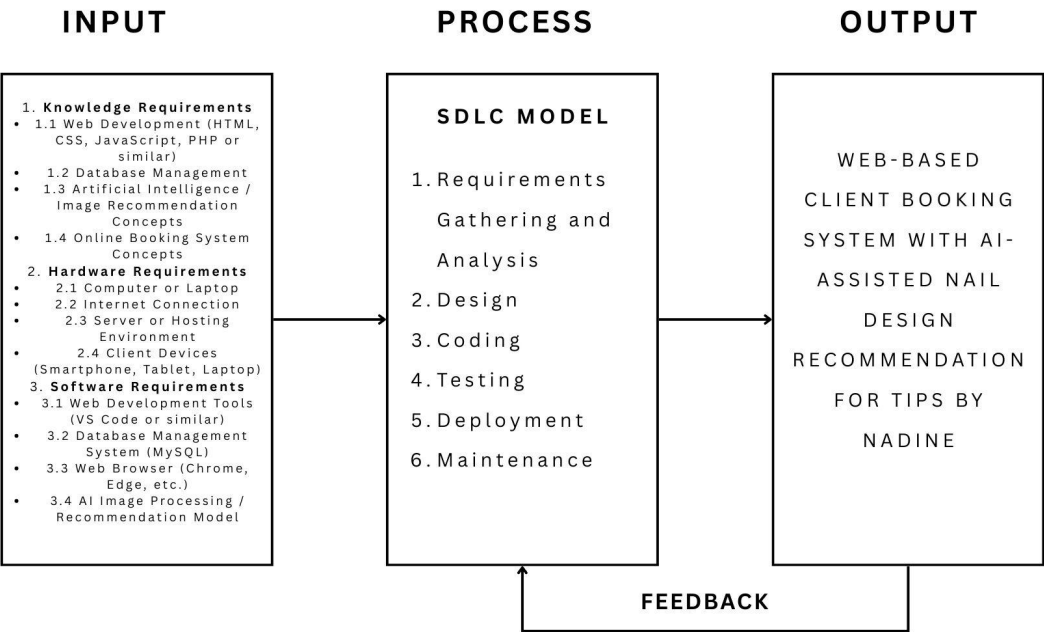


Figure1. Conceptual Framework

This study categorizes the input into three key areas to effectively design and implement the proposed system. The first input is the knowledge requirements, which include an understanding of web development such as HTML, CSS, JavaScript, and PHP, as well as database management and artificial intelligence concepts related to image-based recommendation systems. Knowledge of online booking systems is also important to understand how appointments, scheduling, and client interactions will be managed within the

platform.

The second input is the hardware requirements, which include the devices needed to develop and operate the system. These consist of a computer or laptop used by the developers, a stable internet connection for system access, a server or hosting environment where the system will be deployed, and client devices such as smartphones, tablets, or laptops that will allow users to access the system through a web browser.

The third input is the software requirements, which provide the tools necessary to develop and maintain the system. Programming languages such as PHP, HTML, CSS, and JavaScript are used to build the system interface and backend functions. Web development tools such as Visual Studio Code are used for coding and debugging, while MySQL is used for database management and storage of client records, bookings, and nail design information. Web browsers such as Google Chrome or Microsoft Edge are also required to run and test the system. In addition, an AI image processing or recommendation model is integrated into the system to generate nail design suggestions based on the uploaded reference image of the client.

In order to meet the objectives of the study, Artificial Intelligence (AI) is integrated into the system to assist clients in selecting suitable nail designs. The AI-assisted feature analyzes the uploaded nail image and compares it with the available nail design database to generate personalized recommendations. This feature helps simplify the design selection process and improves

communication between clients and the nail service provider.

The development of the system follows the Software Development Life Cycle (SDLC) model, which includes requirements gathering and analysis, system design, coding, testing, deployment, and maintenance. These stages ensure that the system is carefully planned, properly developed, and tested before it is implemented for actual use.

The output of this process is the Web-Based Client Booking System with AI-Assisted Nail Design Recommendation for Tips by Nadine. The system allows clients to check available schedules, book appointments online, browse nail designs with corresponding prices, upload their nail design references, and receive AI-generated design suggestions. It also sends automated email notifications to both the client and the business owner regarding appointment confirmations and reminders.

By automating appointment scheduling and assisting clients in choosing nail designs, the system improves booking efficiency, reduces scheduling conflicts, and enhances the overall service experience for both the business owner and the clients. The system also helps organize booking records, manage schedules, and maintain nail design information in a centralized database, making the overall business operation more efficient and organized.

General Objective

The general objective of this study is to design and develop a web-based client booking system integrated with an AI-assisted nail design recommendation feature to enhance booking efficiency and improve nail design selection for Tips by Nadine.

Specific Objectives

1. To design and develop a web-based booking system that allows clients to check available dates and schedule nail service appointments online.
2. To develop a nail design gallery where clients can view nail designs with corresponding prices and upload their reference images.
3. To integrate an AI-assisted nail design recommendation feature that provides personalized suggestions based on the client's uploaded nail photo.
4. To implement an automated email notification system that sends appointment confirmations and reminders to both the client and the business owner.

SCOPE AND LIMITATIONS

This study focuses on the design and development of a Web-Based Client Booking System with AI-Assisted Nail Design Recommendation for Tips by

Nadine. The system aims to improve appointment scheduling and assist clients in selecting suitable nail designs through digital automation.

The proposed system allows clients to create accounts, log in securely, check available schedules, and book appointments online. It also includes a nail design gallery that displays organized images, categories, descriptions, and corresponding prices. Clients can upload their nail photos to receive AI-assisted design recommendations based on the uploaded image.

The system includes an administrative module that enables the business owner to manage bookings, schedules, and nail design data. Automated email notifications are also integrated to send confirmations and reminders. The system is web-based and accessible through standard web browsers with an internet connection.

This study is limited to the development and implementation of the system for Tips by Nadine only. It is not designed for multi-branch salons or large-scale beauty businesses. The features are focused mainly on booking management and nail design recommendations.

The AI-assisted recommendation feature provides suggestions based on uploaded images and available nail design data. However, the accuracy of the recommendations may vary depending on the quality of the uploaded photo and the available system database.

The system requires a stable internet connection to function properly. Online payment integration, inventory management, payroll, and accounting features are not included in this study. The system evaluation is limited to selected respondents and may not represent all possible users.

Significance of the Study

This study aims to design and develop a Web-Based Client Booking System with AI-Assisted Nail Design Recommendation for Tips by Nadine. The proposed system will provide a more efficient and organized method of managing client appointments and nail design selections. Through the use of a digital booking platform, the system will reduce manual processes and improve communication between the business owner and the clients.

The system will provide real-time access to available schedules and nail design options, allowing clients to book appointments more conveniently. It will also help reduce common problems such as delayed responses, double bookings, and miscommunication about nail design preferences. By integrating artificial intelligence, the system will assist clients in selecting suitable nail designs based on their uploaded reference images.

The development of this system contributes to the use of technology in small beauty service businesses by introducing an automated booking and recommendation platform. It demonstrates how digital systems and artificial

intelligence can improve service efficiency, organization, and customer satisfaction in small-scale businesses.

Owner. This study will benefit the business owner by providing a more organized and efficient way of managing appointments, schedules, and nail design records. The system will reduce manual workload, minimize booking errors, and prevent overlapping appointments. It will also help the owner monitor client transactions more effectively and improve overall service quality.

Staff. The proposed system will assist the staff by simplifying the booking and design selection process. Instead of manually responding to messages and managing schedules, staff members can rely on the system to automatically handle bookings and send notifications. This will allow them to focus more on delivering quality nail services and improving customer experience.

Client. Clients will benefit from the convenience of booking appointments online at any time. They will be able to check available schedules, browse nail designs with corresponding prices, upload reference photos, and receive AI-generated design suggestions. The system will improve communication, reduce misunderstandings, and make the overall booking experience faster and more convenient.

Researchers. This study will provide the researchers with valuable experience in system design, web development, and artificial intelligence integration. It will

also enhance their knowledge in developing digital solutions that address real-world business problems.

Future Researchers. This study may serve as a reference for future researchers who plan to develop similar web-based booking systems or AI-assisted recommendation systems. It can provide insights into system design, AI integration, and the application of digital solutions in small beauty businesses.

Definition of Terms

Artificial Intelligence (AI). Refers to the technology integrated into the system that analyzes uploaded nail photos and generates personalized nail design recommendations.

Client. Refers to the customer who books nail service appointments and uses the system to browse designs and receive recommendations.

Database. Refers to the organized collection of client information, nail design data, booking records, and other system-related data stored electronically

Owner. Refers to the business owner of Tips by Nadine who manages appointments, schedules, and nail design information through the system.

Nail Design. Refers to the style, pattern, or decorative appearance applied to the nails, which includes images, categories, descriptions, and prices stored in the system.

Staff. Refers to individuals assisting the owner in providing nail services and managing client appointments.

System. Refers to the Web-Based Client Booking System with AI-Assisted Nail Design Recommendation developed in this study.

Web-based. Refers to a system that is accessible through internet browsers and can be used online without requiring installation on a specific device.